

The Labor Market With Frictions

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I Introduction

We studied the **Walrasian labor market**.

Labor supply is determined by

- wages
- UI benefits (income when not working)
- preferences

Labor demand is the marginal product of labor

Key assumption:

- Wages are fully flexible.

Implications:

- There is no unemployment.
- AD does not affect employment.

I.I Sticky Wages

We now introduce **sticky wages**.

Key implication: AD now affects employment.

Basic intuition:

- Nominal AD rises
- Prices rises
- Wages are sticky
- Real wages fall
- Firms hire more labor

Now the model is suitable for analyzing business cycle frequency events.

1.2 The Idea

The basic idea we want to capture:

Unexpected inflation increases output

- either by increasing labor supply or labor demand
- monetary policy has real effects in the short run
- but they wear off as expectations adjust

Anticipated inflation just increases prices.

- this is why money is neutral in the long run

We can tell that story in various ways

- sticky wages → labor demand story
- sticky price expectations → labor supply story
- sticky prices ...

I.3 The Labor Demand Story

The story in a nutshell

1. Wages are sticky (require time to adjust to shocks)
2. Inflation erodes the real wage.
3. At lower real wages, firms hire more labor.
4. Hence, employment is higher when inflation is higher

This is a good story ... but in the model we use a different one.

2 Labor supply story

We model a simpler version of the story (with similar outcomes).

At the start of the period, workers form price expectations P^e .

Labor supply:

- Workers see W and think the real wage is W/P^e
- How much they want to work is given by $N^s(W/P)$.
- How much they actually work is $N^s(W/P^e)$.

2.1 Labor Demand

For simplicity, labor demand is horizontal

$$W/P = 1/(1 + m)$$

- More general (realistic): downward sloping MPL curve

What are we missing?

- Real wages don't vary over the business cycle.

We call m the markup because

$$P = (1 + m)W$$

Details below ...

2.2 The Role of Price Expectations

If inflation expectations are **correct**:

- workers work as much as they want at the market clearing real wage
- full employment

Unexpected inflation ($P > P^e$) implies high W/P^e .

- Workers think the real wage is high
 - even though it's always $1/(1 + m)$.
- They supply more labor and employment rises.

Unexpected inflation stimulates the economy

- by tricking workers into working too much

2.3 Labor Supply

Labor supply:

$$N^s = \hat{F}(W/P^e, z)$$

z : labor market conditions

- unemployment benefits, taxes, etc

Key: N^s depends on the real wage evaluated at P^e (not P).

We assume that N^s is increasing in W/P^e .

2.4 Labor Demand

In general: MPL is decreasing in N

- Firms hire labor up to the point where $MPL = W/P$

We simplify and assume:

- Output is produced from labor only: $Y = N$
- $MPL = dY/dN = 1$
- Marginal cost $MC = W$

Firms charge a markup m over marginal cost: $P = (1 + m)W$

Labor demand is perfectly elastic at **fixed real wage**

$$W/P = \frac{1}{1 + m}$$

Labor Market Clearing

In general we would set $N^S(W/P) = N^D(W/P)$.

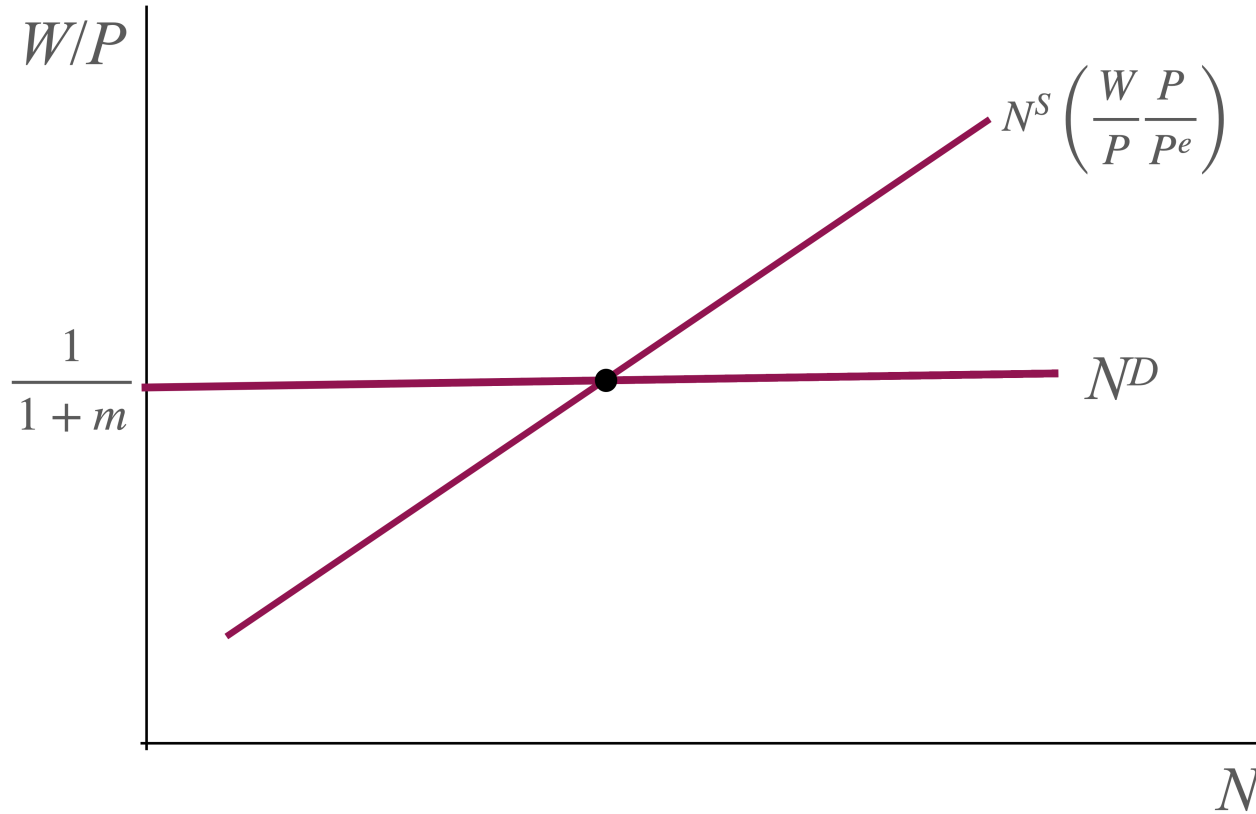
But here N^D is horizontal at the fixed real wage $1/(1+m)$.

So we sub that real wage into labor supply to get market clearing.

$$\begin{aligned} N &= \hat{F}(W/P^e, z) = \hat{F}\left(\frac{P}{P^e} \frac{W}{P}, z\right) \\ &= \hat{F}\left(\underbrace{\frac{P}{P^e}}_{\text{mistake}}, \underbrace{\frac{1}{1+m}}_{\text{real wage}}, z\right) \end{aligned}$$

Employment is increasing in P/P^e and z .

2.5 Labor Market Clearing



For given expectations error P/P^e , labor supply is rising in the real wage.

Model Summary

Production function: $Y = N$

Labor demand:

$$W/P = 1/(1 + m)$$

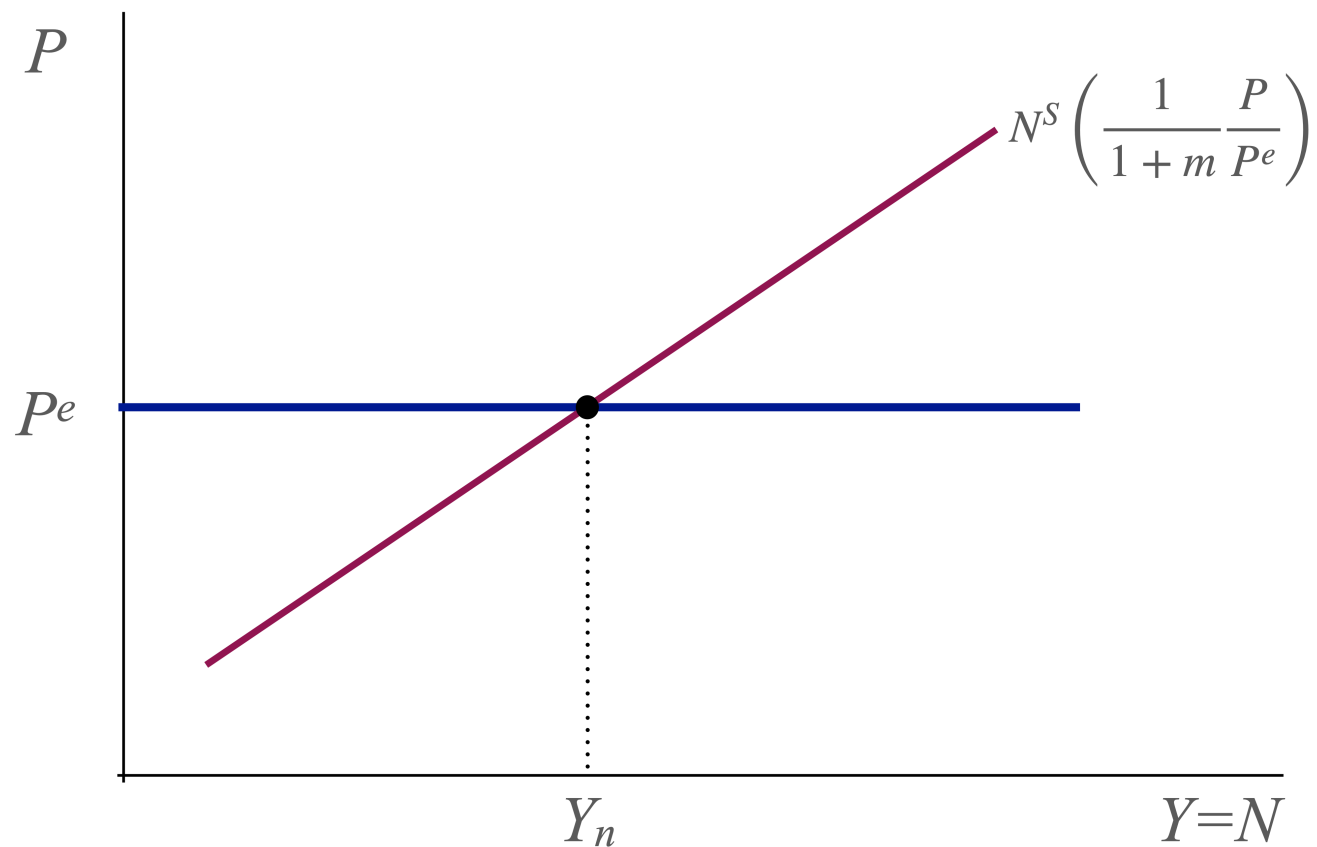
Labor supply:

$$N^S = \hat{F}(W/P^e, z)$$

Labor market clearing:

$$\begin{aligned} Y = N &= \hat{F}(W/P^e, z) \\ &= \hat{F}\left(\frac{P}{P^e} \frac{1}{1 + m}, z\right) \end{aligned}$$

2.6 Summary



Higher (unexpected)
prices



higher employment.

2.7 Intuition

Workers see a high nominal wage and think they see a high real wage.

So they supply more labor.

In reality, price setting by firms fixes the real wage

- Workers are wrong every time.

Until worker's price expectations adjust ($P^e \rightarrow P$), inflation affects employment.

2.8 Exercises

What happens to $Y = N$ when (holding P fixed)

1. price expectations are higher?
2. markups rise?
3. unemployment benefits improve?

2.9 Natural Rate of Unemployment

When price expectations are correct:

$$Y_n = N_n = F\left(\underbrace{\frac{P}{P^e}}_{=1} \frac{1}{1+m}, z\right)$$

This is the medium-run outcome.

- The medium-run supply curve is **vertical**.
- The price level does not matter.

Full employment should really be called “normal employment” or “trend employment.”

- Not everyone works.
- But those who want to work do.

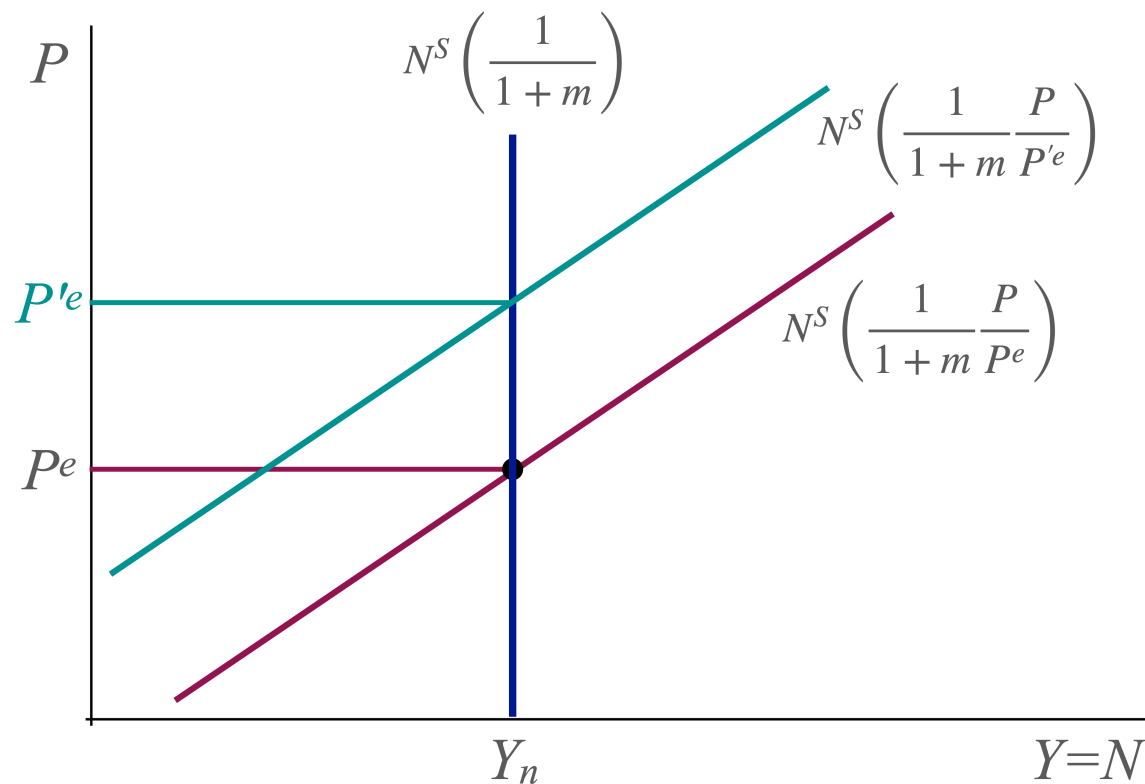
2.10 What affects “full employment?”

$$Y_n = N_n = F\left(\underbrace{\frac{P}{P^e}}_{=1} \frac{1}{1+m}, z\right)$$

Full employment is affected by:

- F
- m
- z

2.11 Long-run Supply Curve



If price expectations eventually catch up with prices ($P^e = P$), we get $Y_n = F\left(\frac{1}{1+m}, z\right)$.

The price level does not matter for employment / output.

2.12 What's Next?

- If price expectations were always correct, we would be done:
 - markups and labor productivity determine the real wage
 - the real wage determines (un)employment
 - employment determines output
- This is what happens in the long run
 - only the supply side matters
- But what happens when $P^e \neq P$?
 - the AS/AD model answers that question

2.13 Does Gov't Spending Create Jobs?

Some questions to discuss...

A bipartisan infrastructure deal ... could create roughly half a million new manufacturing jobs by 2024 ... an analysis conducted on behalf of the trade group Association of Equipment Manufacturers found. ...

[T]he manufacturing jobs would come from \$1.1 trillion spent over eight years ...

CBS New, July 27, 2021

2.14 Destroying Jobs

The same logic applies to measures that raise the cost of doing business:

Michele Bachmann, the congresswoman from Minnesota, in 2011 said she wanted to rename the Environmental Protection Agency “the job-killing organization of America” and Mitt Romney lamented that “Day by day, job-killing regulation by job-killing regulation, bureaucrat by bureaucrat, this president is crushing the dream.”

The Atlantic, Jan 19, 2017

What is the link between regulation and long-run employment?

Blanchard, Macroeconomics, 7th + 8th ed, ch. 7 “The Labor Market”

Further Reading:

- Jones, *Macroeconomics*, ch. 7.